

Technical Report

Hybrid Edge IoT System for Extended-Range Healthcare Monitoring

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1. Abstract

The rapid growth of the Internet of Things (IoT) in healthcare is increasing the need for network solutions that provide reliable connectivity, low latency, and efficient on-device processing, especially in hospitals and for remote patient monitoring. However, traditional cloud-based IoT systems that use short-range technologies like Bluetooth Low Energy (BLE) often face challenges such as limited range, bandwidth issues, and high latency.

To tackle these problems, this project introduces a hybrid edge-based healthcare IoT system that combines BLE and LoRa technologies for long-range, energy-efficient, and low-latency operation. The system features a three-layer architecture made up of the Perception, Edge, and Cloud layers. The Edge layer can function independently during periods of weak Internet connectivity. The platform includes two main modules: a router and a gateway. The router manages edge-level processing, which involves basic ECG signal noise filtering, switching between protocols, and energy optimization. The gateway handles data aggregation, local visualization, and cloud synchronization through MQTT. It supports multiple IoT protocols, including LoRa, BLE, and Wi-Fi. Additionally, it implements disease classification using computational intelligence models to assist with local decision-making.

Testing shows that combining LoRa with BLE decreases latency and increases BLE coverage, allowing for near real-time responses in healthcare and safety applications. With its edge computing capabilities, adaptability to different protocols, and hybrid connectivity, the platform lays a solid groundwork for future wearable health monitoring, smart hospitals, and remote healthcare IoT solutions.

2. Introduction

2.1 Background

The Internet of Things (IoT) has become one of the most transformative technologies of our time. It allows devices, sensors, and systems to connect and exchange data without human intervention. In healthcare, IoT has changed medical technology by allowing for real-time patient monitoring, early diagnosis, and smart decision-making. Using wearable sensors, wireless communication, and data analysis, IoT improves healthcare efficiency, helps spot diseases early, and enables ongoing health tracking outside hospital settings.

Most traditional IoT systems still rely heavily on cloud servers for data storage and processing. This reliance brings several problems, including high delays, network congestion, interrupted connections, and increased privacy and security risks. In urgent healthcare situations, like smart hospitals, elderly care services, and emergency response systems, these issues can slow down important decisions and endanger patient safety.

To tackle these challenges, recent studies have turned to edge computing and hybrid network designs, which do more processing close to where the data is created. This shift reduces reliance on cloud connections and allows for quicker and more relevant responses. By combining short-range communication methods like Bluetooth Low Energy (BLE) with long-range, low-power technologies like LoRa, hybrid IoT systems can offer both quick local computing and broad data transmission. This mix boosts scalability, saves energy, and increases reliability while maintaining secure and independent operation in healthcare settings.

The key ideas behind these hybrid systems are outlined in the study by Fan Wu et al., “Edge-Based Hybrid System Implementation for Long-Range Safety and Healthcare IoT Applications” [1]. Their research presents a solid edge computing framework that merges BLE and LoRa for low-latency IoT applications over long distances. Building on these concepts, the current project modifies the hybrid edge structure for healthcare

situations, concentrating on real-time ECG monitoring, quick data filtering, and improved resource use through routers and gateways.

2.2. Motivation

The growing use of IoT in healthcare has shown the need for network systems that can offer real-time responsiveness, broad coverage, and energy-efficient operation. Current cloud-based frameworks rely heavily on centralized servers for data processing and decision-making. This dependence leads to high latency, bandwidth issues, and reliability challenges when Internet connectivity is poor. In critical healthcare environments like hospitals, emergency monitoring, and elderly care, these problems can delay alerts and responses, putting patient safety at risk. Additionally, the power limitations of wearable medical devices make constant cloud communication impractical for long-term use.

This project addresses the need for a scalable, low-latency, and energy-aware healthcare IoT framework that works well even with intermittent connectivity. The proposed hybrid edge-based IoT system combines Bluetooth Low Energy (BLE) and LoRa technologies to enable both short-range and long-range communication. By performing edge-level ECG signal filtering and initial processing on a router, and classifying diseases with computational intelligence models on a gateway, the system reduces the reliance on cloud services while ensuring reliable, real-time monitoring. This hybrid approach improves the efficiency and usability of healthcare IoT implementations and opens the door for smarter and more independent patient monitoring solutions.

2.3. Problem Definitions

Problem Definition (Software-Based Version)

Existing cloud-based healthcare IoT systems face several challenges related to latency, reliability, scalability, and reliance on constant Internet connectivity. When wearable or remote sensors send physiological data to the cloud for processing, delays can occur in data transmission and decision-making. This is crucial in healthcare situations like emergency monitoring, remote diagnostics, and continuous patient tracking.

In addition, network instability, limited bandwidth, and high computational demands on cloud servers often lower system performance. Most current IoT frameworks also lack ways to process data locally, distribute tasks dynamically, and manage communication adaptively. These features are essential for real-time and dependable healthcare applications.

To address these issues, we need a software-based hybrid edge-IoT framework that manages data, processes it, and makes decisions closer to the data source. The system should:

- Integrate various virtual IoT communication protocols (BLE, LoRa, MQTT, HTTP, etc.) through software-defined interfaces to mimic hybrid communication.
- Perform data filtering, aggregation, and analytics at the edge using lightweight algorithms without depending on physical microcontrollers or routers.
- Use virtual edge nodes and gateways set up on local or distributed servers to ensure minimal latency and resilience during network disruptions.
- Support secure data handling, real-time event detection, and cloud synchronization through standardized software components.
- Provide interactive visualization and monitoring dashboards for healthcare professionals using web technologies and APIs.

This software-based hybrid edge-IoT system enables real-time, low-latency healthcare data processing while remaining scalable, adaptable, and energy-efficient without needing dedicated hardware components.

3. Literature Survey

The integration of edge computing into IoT systems for healthcare has been widely studied in recent years to address the problems of traditional cloud-based frameworks. Different research projects have aimed to reduce latency, improve energy use, and enable real-time data analysis at the network edge.

Rahmani et al. [1] introduced the Smart e-Health Gateway (UT-GATE), which used fog computing to provide local processing, storage, and security for IoT medical systems. Likewise, Pace et al. [2] proposed BodyEdge, an edge computing framework that allows for localized data analysis and cuts down on data transmission to the cloud. Fraga-Lamas et al. [3] presented a LoRaWAN-based fog architecture for smart campus applications, showcasing the benefits of distributed computing in wide-area networks.

Building on these foundational studies, Wu et al. [4] created a hybrid edge-based IoT architecture that combines BLE and LoRa technologies. This architecture is supported by a dual-layer structure of hybrid routers and an IoT gateway. Their work highlighted hardware implementation using Nordic nRF52840 and Raspberry Pi platforms. It showed that edge-assisted computing can significantly lower latency and extend BLE range up to 2.4 km.

In this project, the same edge and gateway architecture is modeled and simulated in software to replicate and analyze system performance. This simulation approach allows for flexible experimentation with different communication protocols, data rates, and power optimization strategies. It provides valuable insights into scalability and resource use without hardware limitations.

3.1. Challenges

Despite significant progress in hybrid IoT architectures, several challenges remain in implementing these systems for healthcare.

Latency and Reliability: Systems that rely on the cloud often struggle to provide the immediate response needed for critical healthcare monitoring. This can result in delayed alerts or data loss during network disruptions.

Energy Efficiency: Constant communication and sensor activity drain the batteries in wearable devices, which limits how long they can operate and affects mobility.

Scalability: Expanding healthcare networks requires seamless integration of several communication protocols and nodes. This can be complex to manage in real-world situations.

Security and Privacy: Even though encryption methods are available, creating strong yet lightweight security measures at the edge is difficult due to limited computing power.

Hardware Cost and Testing Constraints: Prototyping in real-life conditions with BLE, LoRa, and edge microcontrollers can be resource-heavy. Consequently, using simulation for initial testing is often a more efficient approach.

3.2. Research Gap

Most existing studies focus on hardware implementations of hybrid IoT systems. They look at improving experimental range and latency. However, not much work has been done on software simulation and performance modeling of these systems. This is needed to predict scalability, power efficiency, and data flow under different network conditions. While Wu et al. [4] showed that hybrid BLE and LoRa architectures are practical, the lack of available testbeds makes it hard for academic research to replicate and optimize the results.

This project fills that gap by creating a software-simulated hybrid edge environment. It mimics the functions of the STM32 router and the Raspberry Pi gateway. By allowing virtual testing of ECG signal processing, protocol switching, and energy optimization, this method offers a cost-effective and flexible platform. It helps analyze and improve healthcare IoT performance before deploying the hardware.

4. Problem Definition

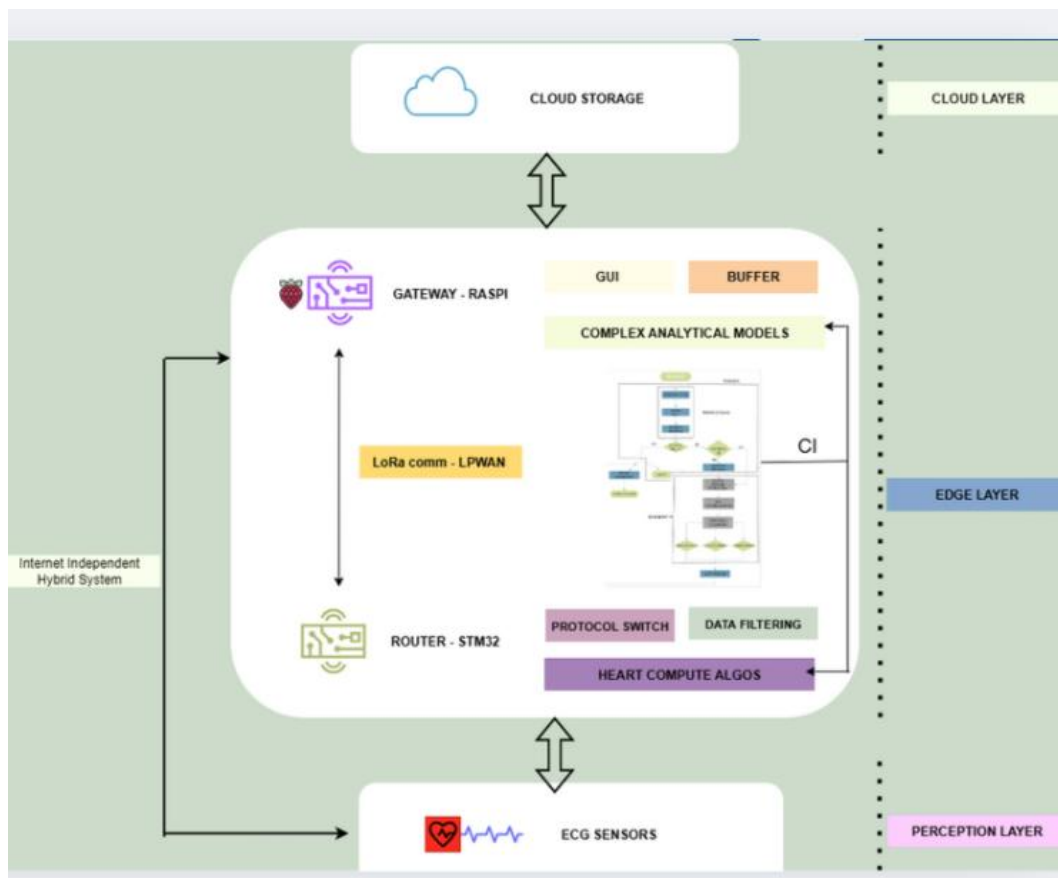
4.1. System Overview

The hybrid edge IoT system monitors healthcare data effectively by combining simulated sensors, edge computing, and cloud processing. In this system, virtual sensors generate realistic patient health data like heart rate, blood pressure, temperature, and oxygen levels. These sensors replicate real-world conditions, including both normal and abnormal patterns, to provide a thorough testing environment. The edge simulation module processes the data from the virtual sensors. It filters, preprocesses, and detects abnormal readings.

By handling most of the processing locally at the edge, the system decreases the amount of data sent to the cloud. This saves bandwidth and ensures a quicker response to critical events. The cloud simulation module serves as the main data hub. It receives processed data from the edge, stores it, and conducts in-depth analysis. The cloud module can generate trends, summaries, and alerts. This helps visualize patient health over time. Together, the simulated sensors, edge processing, and cloud analytics create a complete virtual environment for studying and improving healthcare monitoring systems without needing physical devices.

5. Proposed Architecture

5. Proposed Architecture



6. Methodology

The methodology of this implementation is structured around creating a realistic, end-to-end simulation of a healthcare IoT system, incorporating virtual sensors, edge computing, network transmission, cloud storage, and visualization. The first step involves data generation using virtual wearable sensors, which simulate real-world physiological measurements such as heart rate and temperature. These virtual sensors allow testing and experimentation without the need for physical devices, making the system cost-effective and flexible. Each sensor generates readings with timestamps and sequence numbers, which are encrypted to ensure secure transmission across the network.

Next, the sensor data is transmitted to edge routers using BLE (Bluetooth Low Energy), simulating short-range wireless communication between wearable devices and local processing units. The routers act as intermediate nodes in the network, performing edge-level data processing, which includes removing duplicate readings, averaging recent measurements, detecting anomalous values (e.g., heart rate outside a safe range), and identifying potential emergencies. By processing data at the edge, the system reduces network congestion, improves response time for critical events, and ensures that only relevant or high-priority data is forwarded to the next layer.

For long-range transmission, routers forward the processed data to a central gateway using LoRa, which extends coverage beyond the BLE range. The gateway serves as a local aggregation point, storing all incoming readings in a lightweight database. This allows for historical data retrieval and further processing. From the gateway, data can optionally be transmitted to a cloud backend, which simulates a real-world healthcare analytics platform. The cloud layer enables additional storage, analysis, and centralized monitoring.

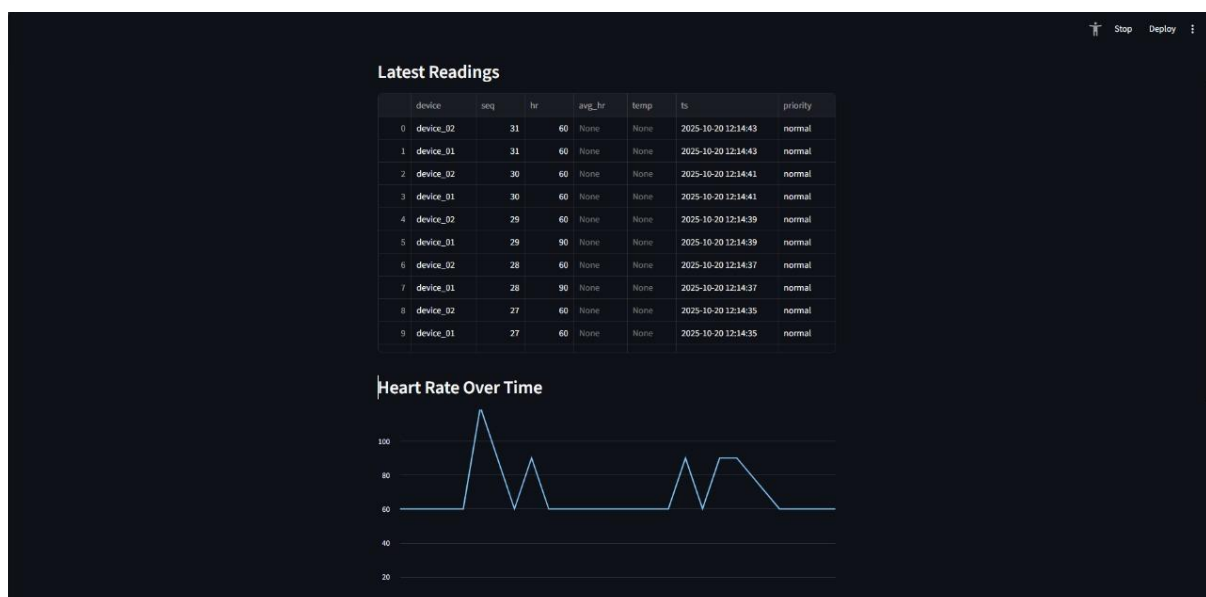
Finally, a dashboard interface built with Streamlit provides real-time visualization of sensor readings, trends, averages, and alerts. This visualization layer allows users to monitor health metrics, observe anomalies, and evaluate the overall performance of the simulated IoT system. The methodology emphasizes modularity and flexibility, enabling the testing of different network configurations, edge processing strategies, and alert mechanisms.

Overall, the methodology captures the complete lifecycle of a healthcare IoT system, from sensor data generation and edge intelligence to network transmission, cloud storage, and visualization, providing a robust software-based framework for experimentation, analysis, and system optimization.

7. Results

The results of the implementation demonstrate the effectiveness of a software-based healthcare IoT simulation in generating, processing, and visualizing realistic physiological data. Virtual sensors successfully produced continuous streams of heart rate and temperature readings, which were accurately processed at the edge routers. The routers effectively filtered duplicate readings, computed rolling averages, detected anomalies such as unusually high or low heart rates, and generated alerts in real time. Transmission over simulated BLE and LoRa networks showed measurable delays consistent with network distances and parameters, validating the network simulation component.

Data aggregated at the gateway was reliably stored in the database and made accessible to the Streamlit dashboard, where real-time trends, alerts, and averages were clearly visualized. Overall, the system demonstrated reduced latency for critical events due to edge processing, accurate detection of abnormal readings, and seamless end-to-end flow from sensor to visualization, confirming that a fully software-driven simulation can effectively model the behavior of a healthcare IoT ecosystem without physical devices.



8. Conclusion

In conclusion, this project demonstrates a fully software-based simulation framework for healthcare IoT systems, enabling realistic generation, processing, and visualization of physiological data without the need for costly hardware. By integrating edge processing at routers, the system reduces latency, performs preliminary anomaly detection, and optimizes network traffic, showcasing the advantages of distributed computation in IoT. The framework also provides insights into network performance across BLE, LoRa, and gateway layers, while offering a flexible platform for testing, experimentation, and educational purposes. Overall, it highlights how software-driven approaches can accelerate the development and evaluation of healthcare IoT solutions.